

Radiotherapy for breast cancer

Radiotherapy is a type of treatment that uses high-energy rays called radiation to treat breast cancer cells. It can help to reduce the risk of breast cancer coming back in the area it is given.

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What is radiotherapy?

Radiotherapy uses high-energy rays called radiation to treat cancer. It destroys cancer cells in the area where the radiotherapy is given, while doing as little harm as possible to normal cells.

Some normal cells in the area being treated can also be damaged by radiotherapy. This can cause side effects. As the normal cells recover, the side effects usually get better.

Radiotherapy is always carefully planned by a team of experts. They will plan your treatment so it does as little harm as possible to normal cells.

Radiotherapy reduces the risk of [breast cancer](#) coming back in the area it is given to. You usually start radiotherapy 4 to 6 weeks after surgery. Sometimes it may start later.

For example, you may start radiotherapy later if your wound needs more time to heal properly. If you are also having chemotherapy, you have radiotherapy after [chemotherapy](#).

Radiotherapy is a standard treatment for breast cancer. But it may be offered as part of a [clinical trial](#).

Radiotherapy after breast-conserving surgery

If you have [breast-conserving surgery](#), your cancer doctor will usually recommend you have radiotherapy to the breast afterwards.

Some people have a very low risk of cancer coming back in the breast after surgery. If you are in this situation, your cancer doctor may talk to you about not having radiotherapy.

Before you decide, you should talk about it carefully with your cancer doctor and nurse. You need to fully understand the advantages and possible risks of not having radiotherapy.

Radiotherapy after a mastectomy

You may still need radiotherapy to the chest after [a mastectomy](#). This will depend on the risk of the cancer coming back in that area. You are more likely to have radiotherapy if:

- the cancer was large
- the cancer had spread to the lymph nodes in the armpit
- there were cancer cells close to the edge of the removed breast tissue.

Radiotherapy to the lymph nodes

If the surgeon removed some lymph nodes from your armpit and these contained cancer cells, you may have radiotherapy to the rest of the lymph nodes.

You may also have radiotherapy to the lymph nodes above the collarbone and behind the breastbone.

Having radiotherapy

You have radiotherapy as an outpatient. It is usually given using equipment that looks like a large x-ray machine. This is called [external beam radiotherapy \(EBRT\)](#), or external radiotherapy. The person who operates the machine is called a therapy radiographer. They will give you information and support during your treatment.

You usually have radiotherapy as a series of short daily treatments. Each treatment is called a fraction or a session. You usually have 5 sessions over 1 week. Sometimes you have 15 sessions over 3 weeks. Your cancer doctor will tell you how many sessions you will need.

If you had breast-conserving surgery, you may have extra radiotherapy to the area where the cancer was. This is called a radiotherapy boost.

Sometimes the boost is given at the same time as radiotherapy to the rest of the breast. Or it may be given at the end of the treatments. This means you may need a few more treatments after finishing your main course of radiotherapy.

If you have radiotherapy to your left side, you will usually be asked to take a deep breath and hold it briefly. This is called deep inspiration breath hold (DIBH). You do this at each of your planning and treatment sessions. DIBH helps protect your heart during radiotherapy treatment to your left side.

Your heart is on the left side of your chest. DIBH moves the heart away from the area being treated. It also keeps you still and reduces the risk of late effects. The website respire.org.uk explains more about DIBH. You may have intensity-modulated radiotherapy (IMRT). This is another type of external radiotherapy. It shapes the radiotherapy beams and allows the radiographer to give different doses of radiotherapy to different areas. This means you have lower doses of radiotherapy to healthy tissue surrounding the tumour.

External radiotherapy does not make you radioactive. After treatment, it is safe for you to be with other people, including children.

In this video, Naman, a Therapeutic Radiographer, explains how radiotherapy is used to treat breast cancer and what people can expect during treatment.



Planning your radiotherapy treatment

You will have a hospital appointment to plan your treatment. You will usually have a [CT scan](#) of the area to be treated. During the scan, you need to lie in the position that you will be in for your radiotherapy treatment.

Your radiotherapy team use information from this scan to plan:

- the dose of radiotherapy
- the area to be treated.

You may have some small, permanent markings made on your skin. The marks are about the size of a pinpoint. They are made in the same way as a tattoo. The marks help the radiographer make sure you are in the correct position for each session of radiotherapy.

These marks will only be made with your permission. Tell the radiographer if you are worried about them or already have a tattoo in the area to be treated.

Treatment sessions

Your radiographer will explain what happens during treatment. At the beginning of each session, they make sure you are in the correct position. Usually, you lie with your arms above your head. If your muscles and shoulder feel stiff or painful, a physiotherapist can show you exercises that may help.

When you are in the correct position, the radiographer leaves the room and the treatment starts. The treatment itself is not painful, and it only takes a few minutes.

The radiographers can see and hear you from outside the room. You can usually talk to them through an intercom, if you need to.

During treatment, the radiotherapy machine may stop and move into a new position. This is so you can have radiotherapy from different directions to the same breast.

If your muscles and shoulder feel stiff or painful, a physiotherapist can show you exercises that may help. Your cancer team can refer you to a physiotherapist.

Radiotherapy to part of the breast

Less commonly, you may be offered radiotherapy to part of the breast instead of the whole breast. This way of giving radiotherapy is usually only given to people having breast conserving surgery.

Men only have a small amount of breast tissue and usually have a mastectomy.

Your cancer doctor or nurse will explain if this is an option for you. They will tell you what the possible side effects are, and any risks involved. They can explain how these treatments compare with external radiotherapy. It is important to have information about all your treatment options.

Radiotherapy to part of the breast can be given in different ways:

Partial breast radiotherapy



Radiotherapy can be given to a smaller area of the breast using external radiotherapy. This is similar to whole-breast radiotherapy. But it only treats where the lump was removed and a small area of normal tissue around it.

Internal radiotherapy



You may have radiotherapy from inside the body (internally) instead of to the whole breast. This is called brachytherapy. It is given over a shorter time.

Hollow tubes are put into the area where the cancer was removed from. Radioactive material is placed into the tubes. The radioactive material may be left in place for a few days. This means you have to stay in hospital. Or you may have it over a few sessions as an outpatient. The radioactive material is removed each time before you go home.

Intraoperative radiotherapy



This type of radiotherapy is also given internally, but during breast-conserving surgery. After removing the cancer, your cancer doctor gives a single dose of radiotherapy to the same area. They give the radiotherapy using a special machine.

After intraoperative radiotherapy, you will not usually need any external radiotherapy to the rest of the breast. But sometimes you may need a short course.

Intraoperative radiotherapy is not suitable for everyone. It is not widely available on the NHS. The National Institute for Health and Care Excellence (NICE) has approved its use, but not as a standard treatment. NICE only covers England and Wales. It should only be used in hospitals that already have these types of machines.

Side effects of radiotherapy for breast cancer

Radiotherapy can cause side effects in the area of your body that is being treated. You may also have some general [side effects](#), such as feeling tired. Sometimes side effects get worse for a time during and after you have finished radiotherapy before they get better.

If you are having the radiotherapy over 1 week, sometimes the side effects may not start for 2 to 3 weeks after treatment.

Your cancer doctor, breast care nurse or radiographer will tell you what to expect. They will give you advice on what you can do to manage side effects. If you have any new side effects or if side effects get worse, tell them straight away.

Skin irritation

If you have white or pale skin, the treated area may get red, dry and itchy. If you have black or brown skin, the treated area may get darker, dry and itchy.

Your nurse or radiographer will give you advice on looking after your skin. If it becomes sore and flaky, your doctor can prescribe creams or dressings to help this.

Skin reactions usually get worse after treatment for a few weeks. But they slowly start to improve 2 weeks after radiotherapy ends.

Here are some tips for skin reactions:

- Do not put anything on your skin in the treated area without checking with your cancer doctor, nurse or radiographer.
- Have cool or warm showers or baths. Turn away from shower spray to protect the treated area.
- Avoid shaving, waxing or using epilators on your underarm on the affected side.
- Gently pat the area dry with a soft towel – do not rub.
- Wear loose clothing. This is less likely to irritate your skin.
- Avoiding swimming if your skin is irritated.

You need to avoid exposing the treated area to the sun during radiotherapy and after treatment finishes. Use suncream with a sun protection factor (SPF) of at least 30.

Tiredness

This is a common side effect that may last for a few weeks or months after treatment. Studies show that exercise can help to manage tiredness caused by treatment. Try to get enough rest and pace yourself. But it is important to balance this with some [physical activity](#), such as going for short walks. This can give you more energy.

Aches and swelling

You may have a dull ache in the treated area. Or you may have shooting pains that last for a few seconds or minutes. You may also notice that the area becomes swollen. These effects may improve after treatment. But you may still have some aches and pains in the area after treatment ends. The area can sometimes stay a little swollen after treatment.

Late effects of radiotherapy

Radiotherapy to the breast may cause side effects that happen months or years after radiotherapy. These are called late effects. Newer ways of having radiotherapy are helping reduce the risk of late effects. If you are worried about late effects, talk to your cancer doctor, breast care nurse or radiographer.

The most common late effect is a change in how the breast or chest area looks and feels.

Radiotherapy can damage small blood vessels in the skin. This can cause red, spidery marks to show. These are called telangiectasia. They may be more common if you had boost doses of radiotherapy.

After radiotherapy, your breast may feel firmer and shrink slightly in size. If your breast is noticeably smaller, you can have surgery to reduce the size of your other breast. If you had [breast reconstruction](#) using an implant before radiotherapy, you may need to have the implant replaced.

You may find the treated area sore or uncomfortable for some time. This usually improves over years. It is not uncommon to get pain in the muscle or ribs at the edge of the breast if you overdo things. Very rarely, radiotherapy may cause lung problems or problems with the ribs.

If you have radiotherapy to the left breast, very rarely it can cause heart problems. Tell your cancer doctor, nurse or radiographer if you notice any problems with your breathing, or have any pain in the chest area. We have more information about the [late effects of breast cancer treatment](#).

About our information

References



Below is a sample of the sources used in our breast cancer information. If you would like more information about the sources we use, please contact us at cancerinformationteam@macmillan.org.uk

ESMO. Early breast cancer clinical practice guidelines for diagnosis, treatment and follow-up. 2019, Vol 30, pp1192–1220. Available from: <https://www.esmo.org/guidelines/guidelines-by-topic/breast-cancer/early-breast-cancer> [accessed 2023].

National Institute for Health and Care Excellence (NICE). Early and locally advanced breast cancer: diagnosis and management. 2018. Updated 2023. Available from: <https://www.nice.org.uk/guidance/ng101> [accessed 2023].

Reviewers



This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer. It has been approved by [Dr Rebecca Roylance](#), Consultant Medical Oncologist and [Professor Mike Dixon](#), Professor of Surgery and Consultant Breast Surgeon.

Our cancer information has been awarded the [PIF TICK](#). Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
-

use illustrations to explain text

- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

Date reviewed

Reviewed: 01 October 2023 | Next review: 01 October 2026

⚠ This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

How we can help

Clinical Information Nurse Specialists

Our Cancer Information Nurse Specialists are dedicated cancer nurses available to talk to on our Macmillan Cancer Support Line.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help